Data to Action: Strengthening Local Climate and Health Programs Through Integrated Analytics and Organizational Planning

Poster Supplementary, NYSPHC Summit 2026

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1 GIS Analysis of Monroe County Flood Hazard Overlays and Disadvantaged Community Intersections for Equity Planning

1.1 Developed on ROC Data

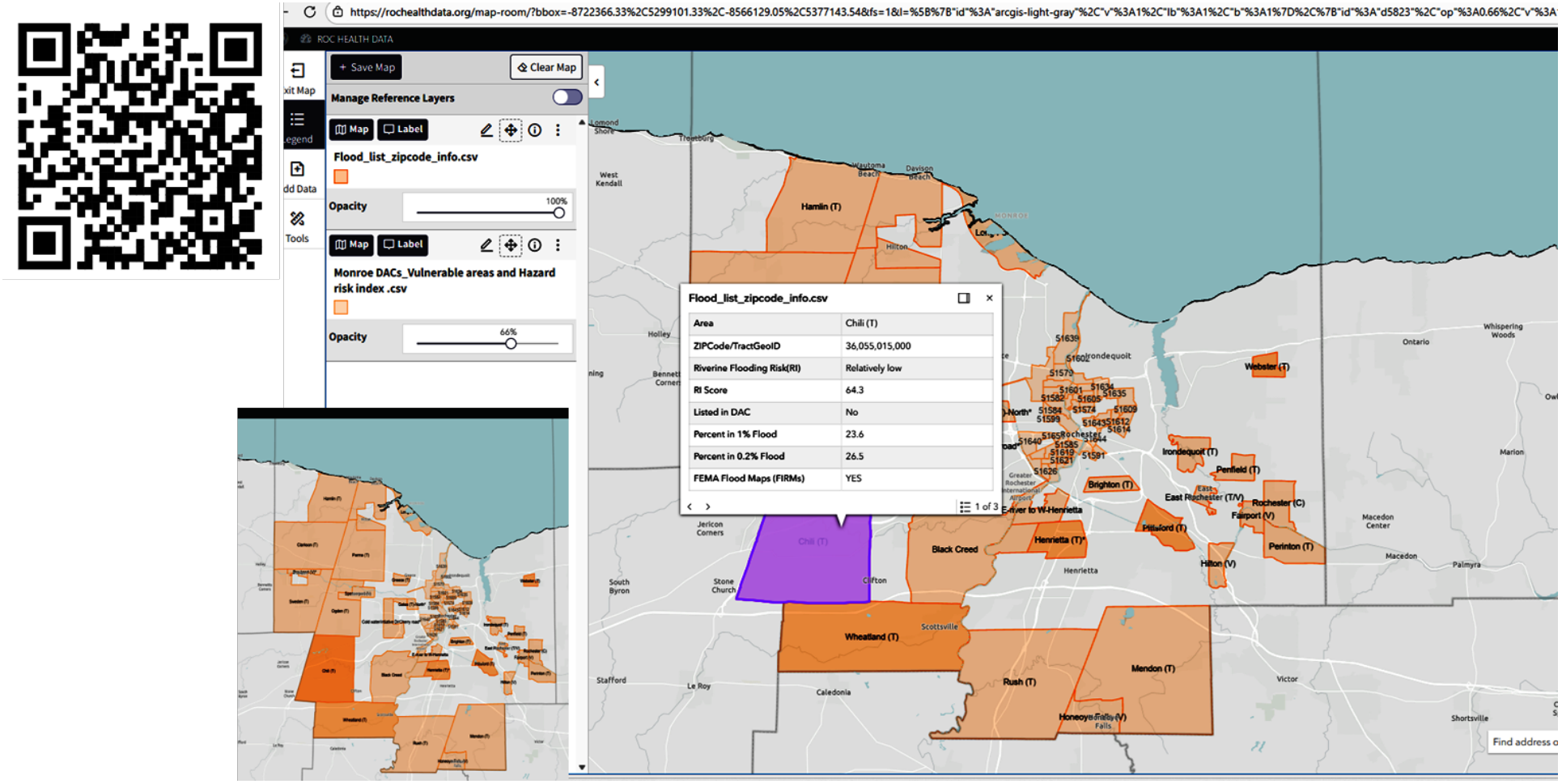


Figure 1: flood hazard overlays-1% & 0.2% floodplains and assessed intersections with county DACs

2 GIS maps of community resources on ROC Health Data tools, including food pantries, shelters and low-income clinics

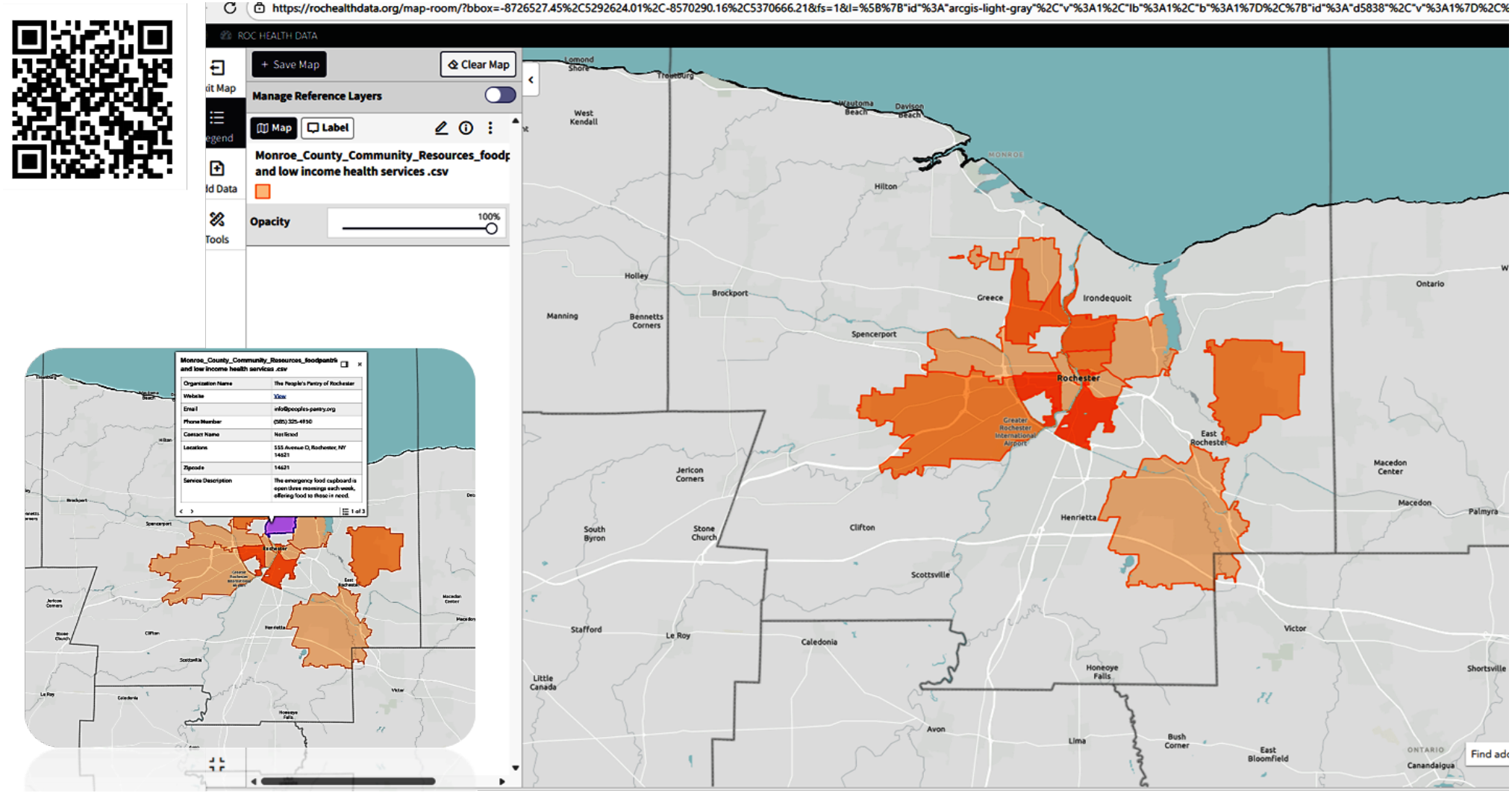


Figure 2: GIS maps tool to connect Climate-Health vulnerable population to community resources

3 Vector Surveillance: Mosquito Seasonal Temperature Thresholds in Monroe County (1970–2024)

3.1 Objective

To evaluate temperature thresholds and environmental conditions that support vector activity in Monroe County in order to inform environmental health monitoring and strengthen vector disease prevention planning.

3.2 Hypothesis

If temperatures remain above 50^0F for 7 consecutive days, mosquitoes will become consistently active, classified as "Active" and $< 50^0F$ classified as "Inactive. The temperature between $50^0F - 86^0F$ are only counted as temperatures above 86°F can disrupt or terminate mosquito life cycles.

3.3 Biological Context

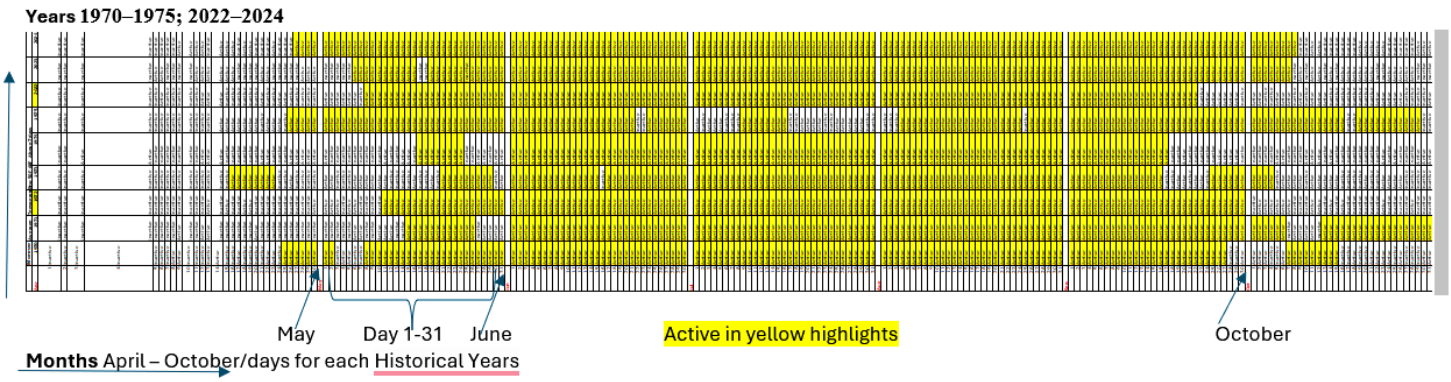
Pupal stage is complete 2–7-days, 14-day life cycle (range 4 days–1 month), with warm, stable temperatures enabling overlapping generations.

3.4 Procedure

Historical Data Analysis: Daily mosquito activity records from 1970–1975 and 2022–2024 were analyzed to identify seasonal patterns.

- Determined the first $\geq$ 7-consecutive-day active streak in spring (seasonal onset).
- Determined the first and last $\geq$ 7-consecutive-day active streaks to identify seasonal onset in spring and seasonal end in fall.
- Measured length of uninterrupted active streaks and compared seasonal duration and consistency across years to assess long-term trends

3.5 Excel Work Flow Screenshot



3.6 Results and Summaries

Month	Avg. Active Days	Activity Pattern	Supporting Interpretation
April	12	Mostly inactive with scattered early streaks	Early possible activity; adult mosquito lifespan typically 1–4 weeks; frequent interruptions observed in earlier decades
May	28	Multiple sustained streaks; longest 11 days (May 11–21)	Environmental conditions increasingly favorable; activity becomes more persistent
June	30	Fully active; uninterrupted activity	Peak stabilization; overlapping generations aligned with 14 day mosquito life cycle
July	30	Peak season; overlapping generations	High relative humidity (72%) and precipitation (84 mm) support sustained activity
August	31	Fully active; consistently high activity	Post rainfall increases observed within 1–2 weeks; maximal seasonal activity
September	20	Decline begins; increased inactive days	Early signs of seasonal decline; recent years show longer late September streaks compared to 1970s
October	14	Decreasing activity with sporadic spikes	Transition toward inactivity with intermittent short lived activity events

Table 1: Monthly mosquito activity patterns summarized from multi year observations

4 Biological Implications

1. Expanding mosquito activity window May–October
2. Reduced winter dormancy
3. Longer favorable season supports overlapping generations

5 Tick-Favorable Temperature Streak Analysis (1971–2024)

5.1 Definition of an Active Day

A day was classified as Active if Daily mean temperature $\geq 40^0F$, part of a ≥ 7 consecutive day streak and not an isolated warm spike. This threshold reflects a more extensive biological activation for questing, feeding, reproduction, and molting survival.

5.1.1 Years Overall Average Days With Temperature Streaks Favoring Tick Active Season

$$\text{Monthly Average Active Days} = \frac{\text{Total Active Days in Month Across All Years}}{\text{Number of Years}}$$

Table 1: Tick Activity (Historical Data 1971–2024)

Month	Avg Active Days	Notes
March	18	Mostly inactive
April	28	Strong transition month
May	30	Fully active
June	31	Peak season
July	31	Consistently high activity
August	31	Consistently high activity
September	30	Consistently high activity
October	31	Consistently high activity
November	24	Declining activity
December	12	Declining activity

5.2 Period Comparison

Table 2: Monthly Average Active Days (1971–2024)

Period	Avg Active Days	Notes
Early Period (1971–1981)	253	Frequent early March activation; shorter warm season overall
Recent Period (2008–2024)	270	Tick favorable season increased about 17 days which equals more than two weeks

Period Summary

- Variability: Min 242 days (1971), Max 282 days (2015), Range = 40 days
- Trend: Monotonic upward; no collapse in recent decades; highest values observed in recent years
- Early spring activation now occurs 1–2 weeks earlier than in the early period

5.3 Biological Implications

Analysis of Monroe County temperature data (1971–2024) using biologically relevant thresholds indicates that ticks experience environmentally favorable conditions for 266 days/year (73% of the year). This extended window supports strong population activity from May through October. Biologically, this suggests:

- An expanding tick season window
- Reduced winter dormancy
- Favorable environmental conditions for population activity primarily between May and October

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